

Jonas Lim Hong Xiang

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EDUCATION & AWARDS

Master of Computer Science (Data Science & AI)

Aug 2024 - Jun 2026

University of Sydney, Australia

Distinction

Singapore IMDA SG Digital Scholar (Infocomm Media Development Authority)

- Capstone Project: AI-Powered Tour Guide using Unitree Go2 quadruped robot.
- ELEC5514 IoT Wireless Sensing & Network: Built real-time chat on STM32 MCU and LoRa transceiver.
- COMP5216 Mobile Computing: Built a food wastage app with TypeScript and React Native.
- COMP5046 Natural Language Processing: Built encoder-decoder models and explored LLM prompting.
- COMP5427 Usability Engineering: Analyzed, designed, and evaluated an e-book.
- COMP5425 Multimedia Retrieval: Built a RAG-based chatbot.
- COMP5426 Parallel & Distributed Programming: Built C programs with pthreads, OpenMP, OpenMPI, and CUDA.
- COMP5318 Machine Learning: Implemented Vision Transformer (ViT) from scratch for age prediction.

Bachelor of Engineering (Aerospace)

Aug 2015 - May 2019

Nanyang Technological University, Singapore

First Class Honors, Highest Distinction

PROFESSIONAL EXPERIENCE

Co-Founder & Software Architect

May 2024 - Present

NockHub, Singapore | nockhub.com

- Built NockHub from scratch, owning delivery from discovery to production launch.
- Gathered operator requirements and turned pain points into roadmap features.
- Designed Figma prototypes and converted workflows into production system architecture.
- Architected a fully serverless backend on AWS (Python, Terraform) with 40+ schemas and 30+ production APIs.
- Shipped 5 cross-platform apps (Flutter) and led 3 engineers on delivery standards.
- Scaled adoption to 4 ranges, 7,000+ users, and 4,000+ bookings per month.

Senior Machine Learning Engineer

Jun 2022 - Jul 2024

Carro, Singapore

- Delivered ML systems across predictive pricing, vehicle damage detection, and engine acoustics.
- Built scalable ML microservices on GPU clusters serving 12,000+ requests per day.
- Replaced Mask R-CNN with YOLOv5 in production; reduced inference time from 500ms to 50ms.
- Built data prep pipelines, feature store, and metadata store; improved training data quality by 50%.
- Partnered with product, operations, and data teams across Singapore, Malaysia, Indonesia, and Taiwan.
- Optimized AWS networking, databases, and training servers; saved USD 120k in 2023.
- Replaced monolithic crawler with distributed architecture; reduced processing from 7 days to 9 hours.
- Reworked team ML infrastructure and workflows to save 75 man-hours per month per engineer.

Data Scientist

Jan 2021 - May 2022

Micron Technology, Singapore

- Optimized SQL in model data pipelines and reduced query time by 50%.
- Built and deployed a predictive yield model to on-premises servers, contributing to USD 11M savings.
- Delivered end-to-end ML solutions from problem framing to production deployment.

Process Engineer

Jun 2019 - Jan 2021

Micron Technology, Singapore

- Lead defect yield improvement initiatives for 120s part types
- Built a defect classification model that improved inspection tool capacity by 10%.

TECHNICAL EXPERTISE

Languages: Python, SQL, TypeScript, Dart, C, Bash
Data Engineering: PySpark
Robotics & Middleware: ROS 2
Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, DocumentDB, RDS

ML/AI: PyTorch, TensorFlow
Backend & APIs: FastAPI, Docker
Cloud & Infrastructure: AWS, GCP, Terraform, Firebase
Frontend / Mobile: Flutter, React Native

HACKATHONS

2025 Global Health Systems Hackathon Hub, "Building High-Value Health Systems: Leveraging AI". Team I Know You Know won 2nd place in Sydney and placed Top 20 globally.

GITHUB PROJECTS

pyspark_project (Data Engineering): PySpark application that aggregates data and publishes to Kafka topics for downstream services.

producer-consumer-ml-system (MLOps): Asynchronous instance-segmentation microservice implemented with Terraform.

CERTIFICATIONS

AWS Certified Solutions Architect - Professional	Jan 2024
Apache Spark 3, Udemy	Dec 2023
AWS Certified Advanced Networking - Specialty	Jul 2023
AWS Certified Solutions Architect - Associate	Aug 2022
Terraform for AWS, Udemy	Jun 2022

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